Advanced Statistical Physics

Lecture Notes

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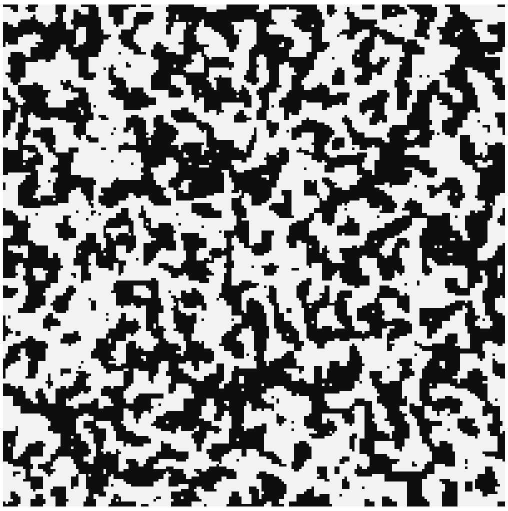

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Lecture 1: Introduction and Van der Waal's Equation of State

What we will discuss here in mainly the equilibrium physics of phase transitions which is both interesting and intriguing. Phase transitions are generally the result of the collective behaviour of interacting particles which produce some change in the thermodynamics of the system. For interaction between particles, we require a *pair potential*, which describes the behaviour of the interaction. One of such potentials is the celebrated *Lennard-Jones* potential (LJP) which goes as:

$$u(r) = \varepsilon_0 \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

where $r = |\mathbf{r}_i - \mathbf{r}_j|$ is the interparticle distance. This is used as a model for intermolecular interactions in many systems. Another "lesser" realistic (but simpler to handle) model can be described by the potential,

$$u(r) = \begin{cases} \infty & r < r_0 \\ -\varepsilon_0 \left(\frac{r_0}{r} \right)^6 & r > r_0 \end{cases}$$

The above potential tells that the minimum possible separation between molecules is r_0, since $u(r)$ blows up below $r = r_0$. Hence the particles act as *hard spheres* (spheres which cannot penetrate each other) of radius $\frac{r_0}{2}$. For $r > r_0$, $u(r) \sim r^{-6}$ which implies *dipolar interactions*.

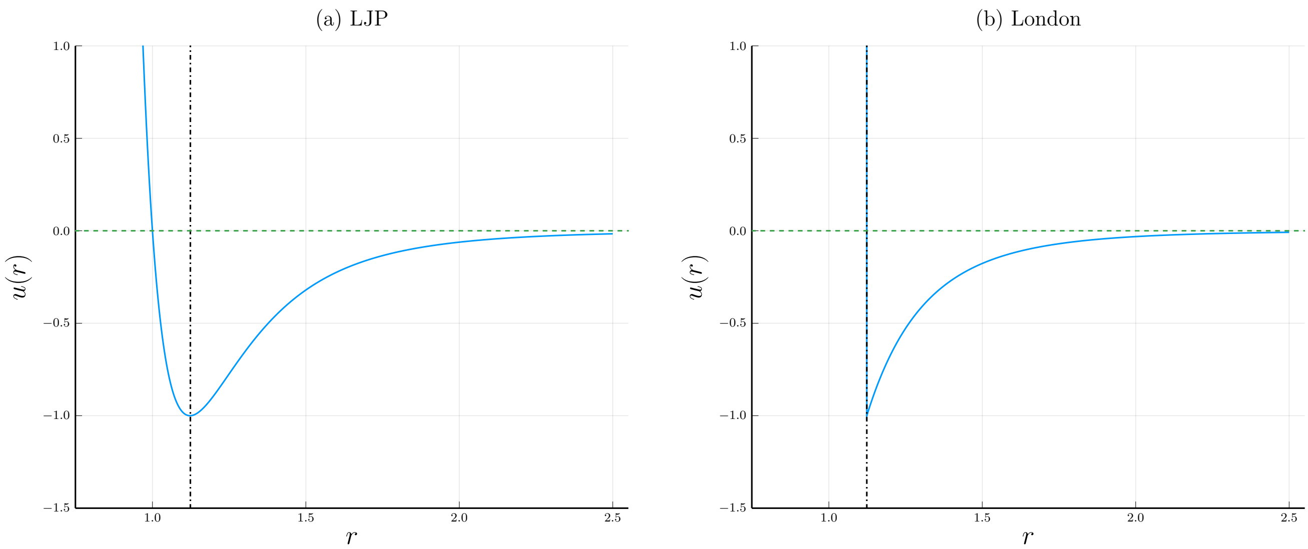


Figure 1: Interaction potentials. (a) Lennard-Jones potential. The vertical line denotes the r at which minima occurs (b) London dispersion potential.

Note that these interactions are mostly *power-law* interactions, $u(r) \sim r^{-s}$. Let d be the dimension of the space in which we are working. Then, it can be shown or motivated that if $s \le d$, then the interaction is long-ranged which is not ideal for us. Extensivity can break down and some other issues (like negative specific heat) may occur in this case. One such example is that of gravitational systems where $u(r) \sim \frac{1}{r}$. If $s > d$ then the interaction is short-ranged. Since $u(r)$ blows up when $r = 0$, in general, it is advisable to impose a hard core repulsion (like in the London force) which ensures that everything stays okay, since interactions act only after $r > r_0$!

1.1. Van der Waal's gas

Let us see an example of this using the Van der Waal's equation which is a correction to the ideal gas equation $PV = Nk_{\mathrm{B}}T$ and successfully explains the condensation process of gases.

We will work in canonical ensemble here. The *canonical partition function* is written as,

$$\mathcal{Z}(N, V, T) = \frac{1}{N!} \int \prod_i \left(\frac{\mathrm{d}^3 p_i \ \mathrm{d}^3 q_i}{h^3} \right) e^{-\beta \mathcal{H}}$$

where $\mathcal{H}$ is the Hamiltonian of the system. The $N!$ factor has been put to include the indistinguishability of the particles and h^3 factor has been put to make the phase space volume dimensionless. The Hamiltonian consists, as usual, of a kinetic and a pair potential part.

$$\mathcal{H} = \sum_i \frac{p_i^2}{2m} + \frac{1}{2}\sum_{i\neq j} u(|\mathbf{q}_i - \mathbf{q}_j|)$$

The momentum integral can be carried out nicely since this is essentially a Gaussian integral. Using independence of the kinetic terms we get,

$$\mathcal{Z}_\mathrm{p} = \left[\int \frac{\mathrm{d}^3 p}{h^3}\ e^{-\beta p^2/2m}\right]^N = \left[\frac{4\pi}{h^3}\int_0^\infty p^2 e^{-\beta p^2/2m}\ \mathrm{d}p\right]^N = \left[\frac{2\pi m k_\mathrm{B} T}{h^2}\right]^{3N/2}$$

Defining the thermal wavelength $\lambda_\mathrm{T} = {h}/{\sqrt{2\pi m k_\mathrm{B} T}}$, we have

$$\mathcal{Z} = \frac{1}{N!\lambda_\mathrm{T}^{3N}}\int \prod_i \mathrm{d}^3 q_i\ e^{-\beta U(\{\mathbf{q}_i\})}$$

where $U(\{\mathbf{q}_i\})$ denotes the interaction term in the Hamiltonian. Tackling this part is extremely hard, since the interactions are between pair of particles, hence independence cannot be assumed.

1.2. Uniform Density Approximation

One of the most popular ways to deal with interactions is through *perturbation*, however, perturbations are generally taken to be *continuous* and hence, non-analyticity (an important aspect of phase transitions) cannot be effectively captured. We will use the method of *mean-field approximation*, which treats the interactions as an "effective potential" which can be found out self-consistently. To simplify our calculations, let us define the number density of the gas, in terms of the delta function,

$$n(\mathbf{q}) = \sum_{i=1}^N\ \delta^{(3)}(\mathbf{q} - \mathbf{q}_i)$$

Then the interaction term becomes,

$$\begin{aligned}
2U &= \sum_i \sum_{j\neq i} u(|\mathbf{q}_i - \mathbf{q}_j|) \\
&= \sum_i \sum_{j\neq i} \int \mathrm{d}q_1\ \delta^{(3)}(\mathbf{q}_1 - \mathbf{q}_i)\cdot u(|\mathbf{q}_1 - \mathbf{q}_j|) \\
&= \sum_i \sum_{j\neq i} \int \mathrm{d}q_1\ \delta^{(3)}(\mathbf{q}_1 - \mathbf{q}_i)\cdot \int \mathrm{d}q_2\ \delta^{(3)}(\mathbf{q}_2 - \mathbf{q}_j)\cdot u(|\mathbf{q}_1 - \mathbf{q}_2|) \\
&= \int \mathrm{d}q_1 \mathrm{d}q_2\ \sum_i \sum_{j\neq i} \delta^{(3)}(\mathbf{q}_1 - \mathbf{q}_i)\cdot\ \delta^{(3)}(\mathbf{q}_2 - \mathbf{q}_j)\cdot u(|\mathbf{q}_1 - \mathbf{q}_2|) \\
&= \int \mathrm{d}q_1 \mathrm{d}q_2\ n(\mathbf{q}_1) n(\mathbf{q}_2) u(|\mathbf{q}_1 - \mathbf{q}_2|)
\end{aligned}$$

We will use a heavily *unmotivated* approximation where we will treat the density to be uniform, independent of the spatial coordinate, that is, $n(\mathbf{r}) \equiv n$ Then,

$$U = \frac{n^2}{2}\int \mathrm{d}q_1^3 \mathrm{d}q_2^3\ u(|\mathbf{q}_1 - \mathbf{q}_2|)$$

Let us define the variables $\mathbf{R} = \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2)$ and $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ from which we have $\mathbf{r}_1 = \mathbf{R} + \frac{\mathbf{r}}{2}$ and $\mathbf{r}_2 = \mathbf{R} - \frac{\mathbf{r}}{2}$. The Jacobian of this transformation can be written as,

$$\mathbf{J} \equiv \frac{\partial(\mathbf{r}_1, \mathbf{r}_2)}{\partial \mathbf{R}, \mathbf{r}} = \begin{pmatrix} \mathbb{1} & \frac{1}{2}\mathbb{1} \\ \mathbb{1} & -\frac{1}{2}\mathbb{1} \end{pmatrix} \implies J = \det(\mathbf{J}) = -1$$

Using the new variables, the integral becomes,

$$U = \frac{n^2}{2}\int \mathrm{d}^3R\ \mathrm{d}^3r\ u(r) = \frac{n^2V}{2}\int_{r_0}^{\infty} \mathrm{d}^3r\ u(r) = \frac{N^2}{2V}\int \mathrm{d}^3r\ u(r)$$

where we have used $n = {}^{N}\!/_{V}$ and r_0 is the hard-core cutoff. Now, assume a power-law interaction for $r > r_0$, $u(r) = -\varepsilon_0\left(\frac{r_0}{r}\right)^s$ using which we get,

$$U = -\frac{\varepsilon_0 N^2}{2V}\cdot r_0^s \int_{r_0}^{\infty} 4\pi r^2 \mathrm{d}r\ \frac{1}{r^s} = -\frac{4\pi\varepsilon_0 N^2 r_0^s}{2V(3-s)}\ r^{3-s}\bigg|_{r_0}^{\infty}$$

As we can see, U is finite only when $3 - s < 0 \implies s > 3$ where 3 was our spatial dimension. Assuming $s > 3$, we have $r^{3-s} \to 0$ and we have

$$U = \frac{4\pi\varepsilon_0 N^2 r_0^s}{2V(3-s)} r_0^{3-s} = -\frac{4\pi\varepsilon_0 N^2 r_0^3}{2V(s-3)} = -\frac{aN^2}{V} \qquad \text{where,}\ \ a \equiv \frac{2\pi\varepsilon_0 r_0^3}{s-3}$$

We get an 'effective' interaction potential using the *uniform density approximation.* From here, the partion function now becomes,

$$\mathcal{Z} = \frac{e^{\beta a N^2/V}}{N!\lambda_{\mathrm{T}}^{3N}}\int\prod \mathrm{d}q_i$$

Now, the integral is simply not equal to V^N, since we have assumed a hard-sphere gas and each particle has some finite volume. Assume that each particle occupies a small volume ω, which cannot be occupied by the other particles. If one particle occupies volume say V, then the second particle can occupy $V - \omega$, the third occupies $V - 2\omega$ and so on. Then the integral will be just,

$$\begin{aligned}
\int\prod \mathrm{d}q_i \equiv \prod_{i=0}^{N-1}(V - i\omega) &\approx \prod (V - j\omega)(V - (N-j)\omega) \\
&= \prod V^2\left(1 - \frac{j\omega}{V}\right)\left(1 - \frac{(N-j)\omega}{V}\right) \\
&\approx V^N \prod\left[1 - \frac{(j + N - j)\omega}{V} + \mathcal{O}(({}^{\omega}\!/_{V})^2)\right] \\
&\approx V^N\left[1 - \frac{N\omega}{V}\right]^{N/2} = \left[V\left(1 - \frac{N\omega}{V}\right)^{1/2}\right]^N
\end{aligned}$$

where we have made the product over $\approx N/2$ pairs. Now, assume $N\omega \ll V$ from which we can use the approximation $(1 + x)^n \approx 1 + nx$ to get,

$$\left[V\left(1 - \frac{N\omega}{V}\right)^{1/2}\right]^N \approx \left[V\left(1 - \frac{N\omega}{2V}\right)\right]^N = \left[V - \frac{N\omega}{2}\right]^N$$

We see that the available volume is reduced by a factor of $N\omega/2$. This was the final step and we can now write the partition function,

$$\boxed{\mathcal{Z}(N, V, T) = \frac{e^{\beta a N^2/V}}{N!\lambda_{\mathrm{T}}^{3N}}\left[V - \frac{N\omega}{2}\right]^N} \tag{1}$$

The free energy can be obtained from the partition function using $F = -k_{\mathrm{B}}T\ln\mathcal{Z}$,

$$F = -\cancel{k_{\mathrm{B}}T}\cancel{\beta}\frac{aN^2}{V} - Nk_{\mathrm{B}}T\ln\left[V - \frac{N\omega}{2}\right] + c(N, T)$$

and from here, we can find the pressure $P = -\left(\frac{\partial F}{\partial V}\right)_{N,T}$

$$P = -\frac{aN^2}{V^2} + \frac{Nk_\mathrm{B}T}{(V - \frac{N\omega}{2})} \implies \boxed{\left(P + \frac{aN^2}{V^2}\right)(V - Nb) = Nk_\mathrm{B}T} \tag{2}$$

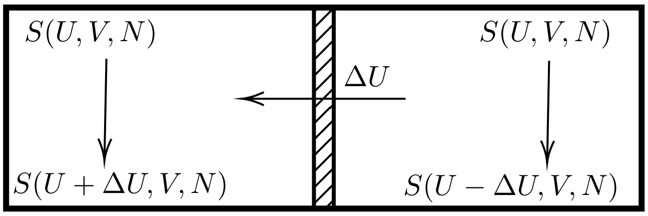
This is the Van der Waal's equation, where we have defined $b = {}^{\omega}/_{2}$. This can be written in an alternate form using the number of moles, $\mathfrak{n} = {}^{N}/_{N_\mathrm{A}}$ where N_A is the Avogadro's number:

$$\left(P + \frac{a'\mathfrak{n}^2}{V^2}\right)(V - \mathfrak{n}b') = \mathfrak{n}RT \iff \left(P + \frac{a'}{v_m^2}\right)(v_m - b') = RT$$

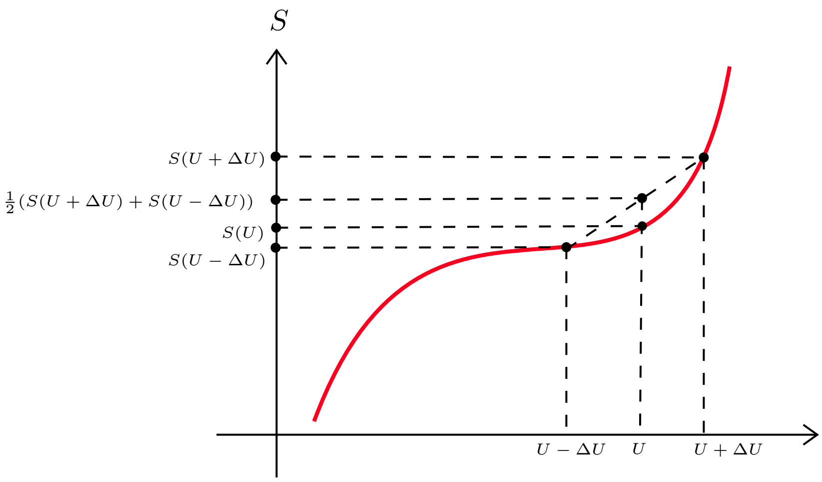
where $a' = N_\mathrm{A}^2 a$ and $b' = bN_\mathrm{A}$ and v_m is the molar volume.

Lecture 2: Stability of Thermodynamic Systems

We will discuss about the conditions under which the system in consideration is stable. This will in turn help us to understand *phase transitions* which are consequences of instability!



Consider two identical subsystems, with entropies $S(U,V,N)$, separated by an adiabatic wall. Suppose we remove an amount of energy ΔU from the first subsystem and transfer it to the second, the total energy changes from $2S(U,V,N)$ to $S(U+\Delta U) + S(U-\Delta U,V,N)$. Suppose that the entropy of the system looked something like in the diagram below.



Then, the resultant entropy after this transfer would be greater than the initial entropy. This would lead to inhomogenities in the system. Within one subsystem, there would be an uneven re-distribution of energy within different regions. This loss of homogenity indicates a *phase separation*!

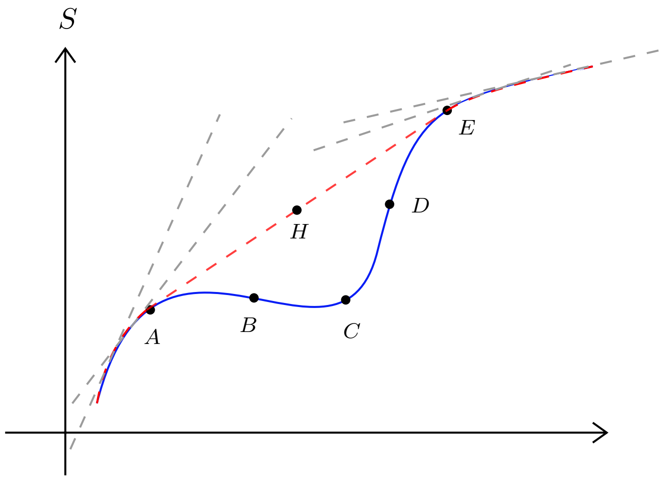
Thus the stability condition is the *concavity of entropy*, that is,

$$S(U-\Delta U,V,N) + S(U+\Delta U,V,N) \leq 2S(U) \tag{3}$$

Taylor expansion of the left hand side leads to the condition (under the limit $\Delta U \to 0$)

$$\boxed{\left(\frac{\partial^2 S}{\partial U^2}\right)_{V,N} \leq 0} \tag{4}$$

This implies that the curvature of the entropy should be downwards! It might happen that we obtain a 'locally convex' entropy curve from some calculation or from experimental data, which does not obey concavity.



This *convex intruder* cannot describe the equilibrium state and we have to obtain the *stable thermodynamic fundamental equation* from this curve. To do that, we construct all the *superior tangents*, that is, tangents which are above this curve. Now, within the family of tangents, there will be one line which will touch the curve at two places.

In the diagram, tangent at any point after A (but before E) will cross the curve. Now, the tangent at A will touch the curve at E. If it had not touched, it would have either intersected the curve or gone above E (If it has intersected the curve, then the tangent at A should not have been considered at all since we are considering only superior tangents. If it had gone over E, this means that there must be some points after A, where this touching will occur. In any case, what we want to say is that, there will be a 'common tangent' at points A and E).

Now, take the envelope of this family of tangents and this will be the fundamental thermodynamic equation that describes the equilibirum states.

Any point on the straight line denotes a phase-separated state, where part of the system is in state A and another part is in E. This happens because of the convex curve of entropy, which creates an inhomogenity.

Note that in the region AB and DE, entropy is still concave, hence these are called *metastable* or locally stable states. These states satisfy Eq.(4) but does not satisfy Eq.(3) while the region BCD satisfy neither condition.

NOTE:

A generic condition for stability is given by $\delta P \delta V \leq 0$ where δP and δV represent a small deviation from the equilibrium value. We can write,

$$0 \leq \delta V \left[\left.\frac{\partial P}{\partial V}\right|_T \delta V + \frac{1}{2} \left.\frac{\partial^2 P}{\partial V^2}\right|_T (\delta V)^2 + \frac{1}{6} \left.\frac{\partial^3 P}{\partial V^3}\right|_T (\delta V)^3 + \dots \right] \tag{5}$$

The first term has $(\delta V)^2$ which is always positive. Hence to make this term negative, we need

$$\left.\frac{\partial P}{\partial V}\right|_T \leq 0$$

Suppose that $\left.\frac{\partial P}{\partial V}\right|_T = 0$. Then the second term has $(\delta V)^3$ and irrespective of what sign $\left.\frac{\partial^2 P}{\partial V^2}\right|_T$ has, δV can always be chosen to make the overall term positive. Hence the only option we have is that

$$\left.\frac{\partial^2 P}{\partial V^2}\right|_T = 0$$

Similar to the first term, we have the third term as,

$$\frac{\partial^3 P}{\partial V^3}\Big|_T \leq 0$$

These define the stability conditions for our thermodynamic system.

2.1. Stability Relations for Potentials

We will now discuss what are the consequence of stability on the thermodynamic potentials, F and G. We will use the fact that for a mechanically stable system, the *specific heat* and the *isothermal compressibility* must be positive for all temperatures. It can be motivated since compressibility and specific heat are related to number and energy fluctuations, σ_{N} and σ_{E} respectively, both of which are positive.

From appendix A, we see that $S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$ from which we get:

$$\left(\frac{\partial^2 F}{\partial T^2}\right)_{V,N} = -\left(\frac{\partial S}{\partial T}\right)_{V,N} = -\frac{1}{T}C_v \leq 0$$

Similarly, we can use $S = -\left(\frac{\partial G}{\partial T}\right)_{P,N}$ from which we get:

$$\left(\frac{\partial^2 G}{\partial T^2}\right)_{P,N} = -\left(\frac{\partial S}{\partial T}\right)_{P,N} = -\frac{1}{T}C_p \leq 0$$

From this, we infer that $G(T, P, N)$ and $F(T, V, N)$ are concave functions of temperature. Apart from temperature, since F depends on V and G depends on P, let us check for pressure and volume too. We note that $V = \left(\frac{\partial G}{\partial P}\right)_{T,N}$ and $P = -\left(\frac{\partial F}{\partial V}\right)_{T,N}$

$$\begin{aligned} \left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N} &= \left(\frac{\partial V}{\partial P}\right)_{T,N} = -V\kappa_{\mathrm{T}} \leq 0 \\ \left(\frac{\partial^2 F}{\partial V^2}\right)_{T,N} &= -\left(\frac{\partial P}{\partial V}\right)_{P,N} = \frac{1}{V\kappa_{\mathrm{T}}} \geq 0 \end{aligned} \tag{6}$$

where we have appropriately used the definitions of isothermal compressibility and heat capacity. Using this, we find that G is a concave function of pressure and F is a convex function of volume. From the above expressions in Eq 6, we can also write

$$\left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N} = -\left\{\left(\frac{\partial^2 F}{\partial V^2}\right)_{T,N}\right\}^{-1} \tag{7}$$

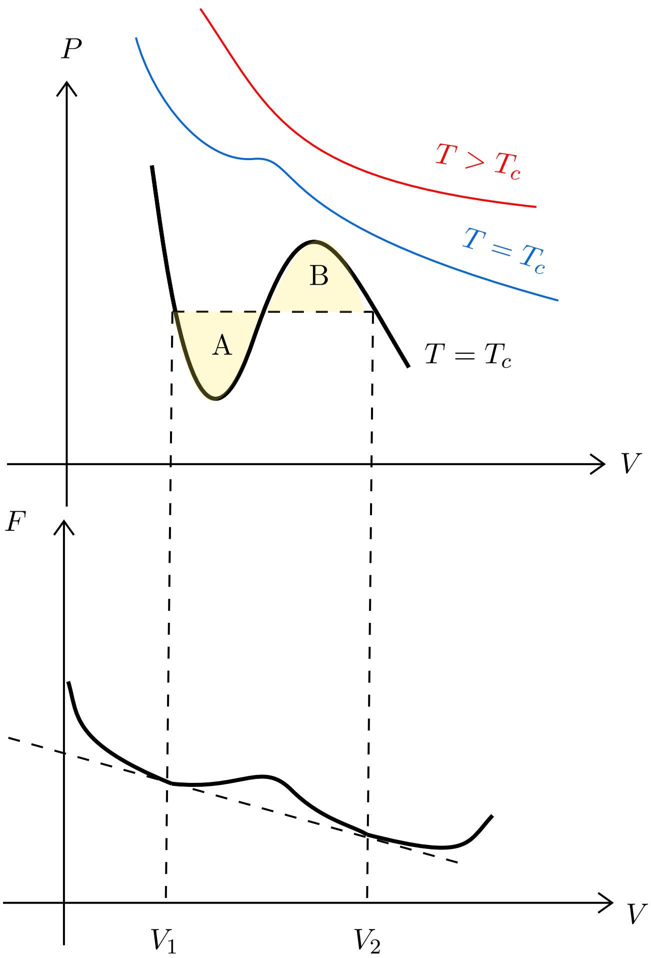
2.2. Stability Analysis of Van der waal's solutions

Recall the Van der Waal's equation from Eq (2),

$$\left(P + \frac{aN^2}{V^2}\right)(V - Nb) = Nk_{\mathrm{B}}T$$

At high temperature, the P vs. V curves at constant temperature (*isotherm*) follows the usual ideal-gas curve. However, there exists a critical temperature T_{c} (which can be found out from the equation), such that, for $T < T_{\mathrm{c}}$, there always exist a region, where the slope $\frac{\partial P}{\partial V}$ is positive. Since $\kappa_{\mathrm{T}} = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{N,T}$, this implies a negative compressibility, which, as motivated before, is *unphysical.*

This leads to a breakdown of the theory and Maxwell subsequently proposed an *ad hoc* remedy, similar in line to the construction of superior tangents that we noted earlier.



Consider the diagram above. Here, for a subcritical temperature, we see a region between V_1 and V_2 where the slope is positive. This region corresponds to the *concave intruder* (since free energy is a convex function of volume) in the free energy plot. Similarly as before, we replaced the concave part with a straight line denoting the commong tangent. This corresponds to joining the region between V_1 and V_2 in the P-V plot with a straight horizontal line. It can also be shown that the horizontal line is at just such a value of the pressure that the areas labelled A and B are equal and hence this construction is also called *equal-area* construction.

Lecture 3: First Order Phase Transition

Let us now see some qualitative aspect of phase transition. Each phase transition can be viewed as a result of the failure of the stability criteria. We will see some characteristic aspects of first order phase transition[1]. However, let us first note some geometric interpretation of the thermodynamic potentials.

3.1. Geometric Interpretation of Potentials

We will consider the relation $G = F + PV$ which can be obtained from the Legendre transform of the internal energy.

[1] In modern language, these are called *abrupt phase transition*

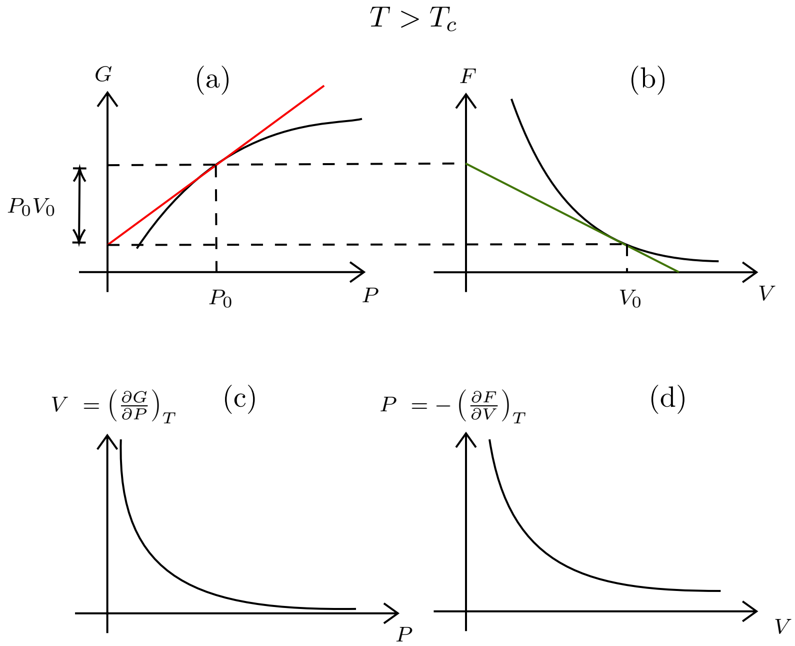


Figure 2: Relation between G and F. (a) Variation of Gibbs potential with pressure at a temperature $T > T_c$ (b) Corresponding plot of Helmholtz potential with V (c) and (d) denotes the P-V diagrams constructed from these potentials

In the above figure, observe the plot of G vs. P, which is smooth. We draw the tangent line at a $P = P_0$ and this intercepts the y-axis somewhere. The equation of the tangent is $G = PV + c$ as $V = \left(\frac{\partial G}{\partial P}\right)_{T,N}$ and it passes through the point (P_0, G_0). Thus we have,

$$G_0 = P_0V_0 + c \implies c = G_0 - P_0V_0$$

The tangent has an intercept of $G_0 - P_0V_0$ and so the remaining length as shown by the arrow in is Fig.2 (a) is $G_0 - (G_0 - P_0V_0) = P_0V_0$. From the length of the intercept of the tangent, we can obtain the Helmholtz potential for this value of P_0. We also obtain the volume from the slope of the tangent. The corresponding F vs. V graph has been plotted using these values. Also, plotted are the V vs. P graph (using the slope of the tangent of the Gibbs potential) and the P vs. V graph (using the slope of the tangent of the Helmholtz potential). This is the case for $T > T_c$. Now let us see the case when $T < T_c$ and the phase transition has occurred. We will see this with respect to the Van der Waal's equation.

$T < T_c$

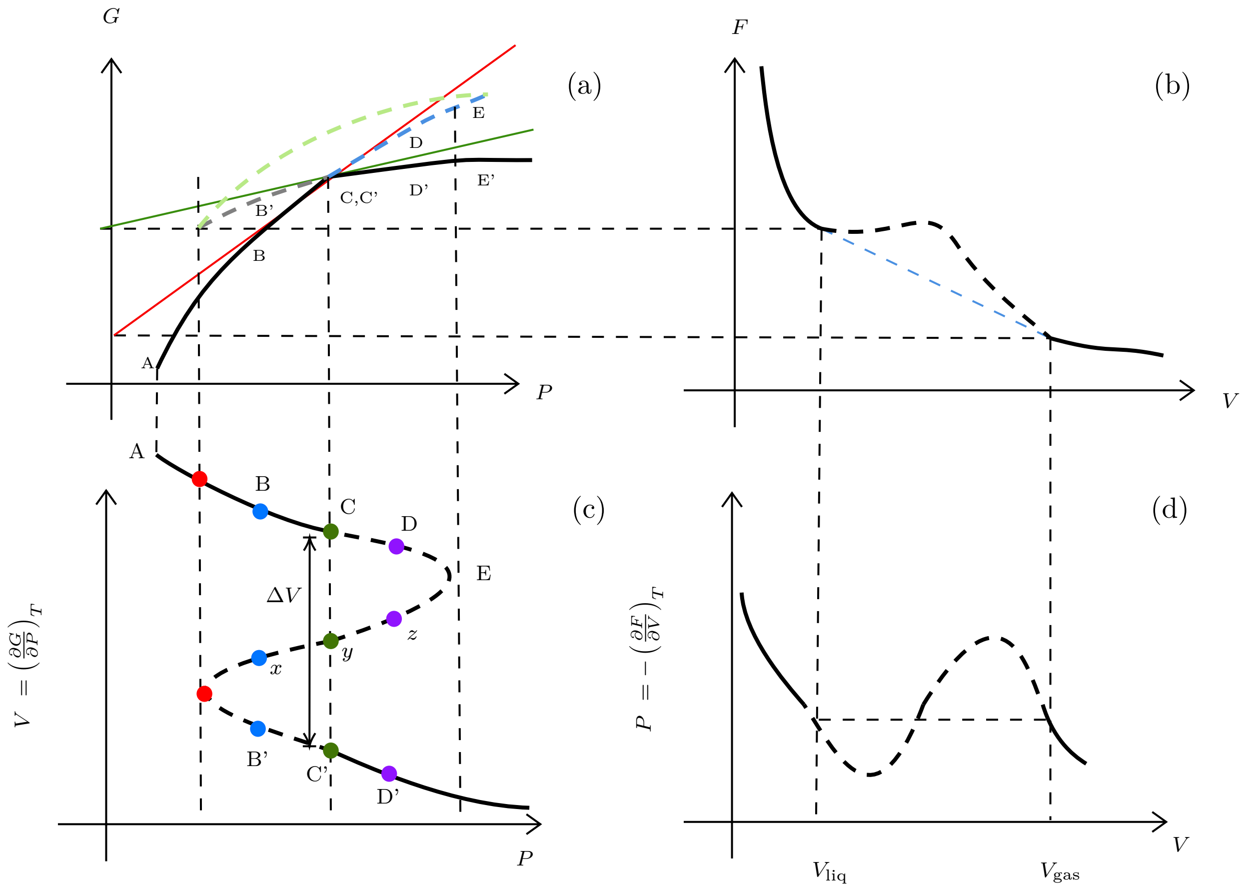


Figure 3: Plot of different quantities for $T < T_c$

In this scenario, G will have a 'kink' for some pressure $P = P_0$, since for first order phase transition, the derivative of the free energy is discontinuous. At this point, $\left(\frac{\partial^2 G}{\partial P^2}\right) \to \infty$ (even though this quantity cannot be defined exactly at $P = P_0$) and using Eq.(7), we get $\left(\frac{\partial A}{\partial V}\right) = 0$. Thus we expect that the corresponding plot of A has a *straight line* portion, as is shown in Fig.3 (b) with a blue dashed line.

Let us now see how the plot of G in (a) is actually constructed. For that, we need the isotherm in (c). Let us start at A and then, integrating we have

$$G(P) - G(P_{\text{A}}) = \int_{P_{\text{A}}}^{P} V \mathrm{d}P$$

and we do this for all values of P. This creates a closed loop kind of plot, since for a single P, we have three values of V. Note that, this disappears for higher temperature since then, Van der Waal's equation consists of only one real and two complex roots.

What happens physically? Let us suppose our system is in a temperature bath and the temperature is fixed. So, consider the isotherm in (c) and start increasing pressure from P_A. The system will follow this isotherm. At P_A, there is a unique volume and thus a unique state of the system. Now, if we increase further, for each pressure, we will get multiple values of volume, so there are different states to choose from.

Consider the blue points (all having same P but different V). Of these, the state x is unstable and hence the system has two states B and B' to choose from. The state chosen is the one with the minimum value of G which is determined from panel (a). As clearly seen, the physical state chosen is B. Increasing pressure further, we get to C where the kink is there.

On increasing further, we see that the state chosen is D' since this has lesser value of G. The blue and gray dashed lines in (a) denote the branch of G with higher value of G and thus, is not chosen. The green dashed line represent the unstable region where the states, say x, y or z lies. We see that in (c), no state from the dashed region is selected and hence after C, there is a volume jump, denoted by ΔV. The physical isotherm is thus the one created from the Maxwell equal area construction (as discussed before) and the straight line in the isotherm denotes the *coexistence* of two phases.

The points C and C' are determined by the condition that $G(P_c) = G(P_{c'})$ and hence, $\int\limits_{P_c}^{P_{c'}} V \mathrm{d}P = 0$, which proves that the areas divided by the line segment are indeed equal.

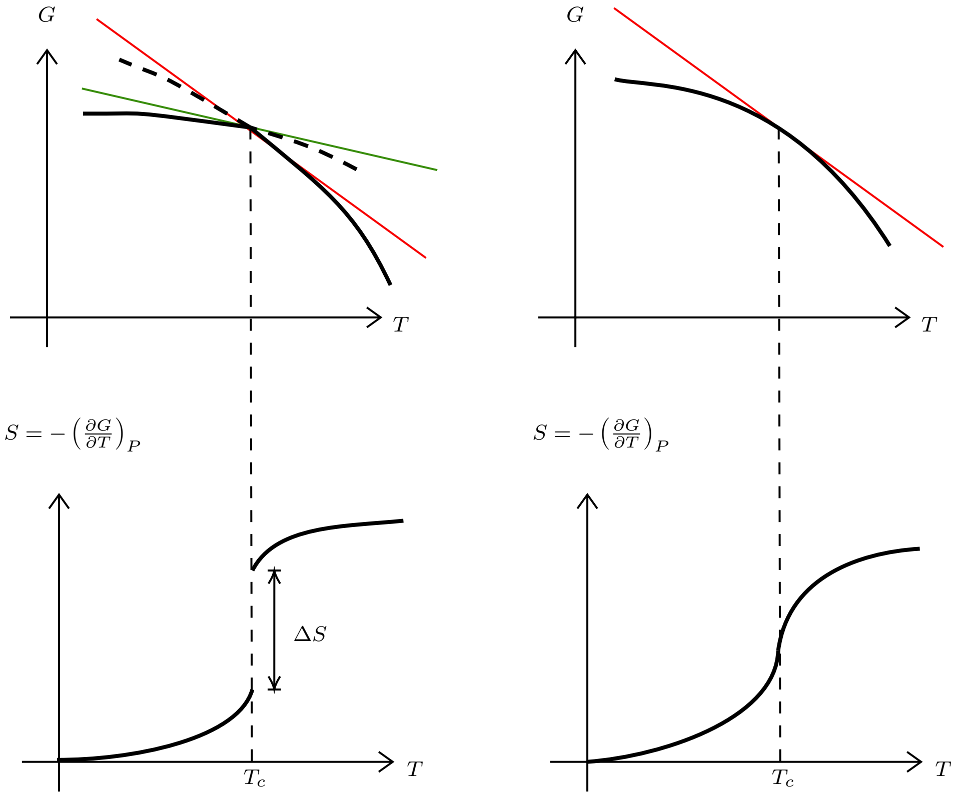


Figure 4: The left column shows the variation of G and S with temperature for first order phase transition, clearly showing an entropy discontinuity. The right column shows this for a higher order phase transition.

There is also an entropy change associated with phase transitions. For first order phase transitions, there is an abrupt entropy jump since entropy is defined as $S = -\left(\frac{\partial G}{\partial T}\right)_P$ and G has a 'kink', making it non-differentiable at the critical point. This entropy jump is associated with the latent heat $L = T_c \Delta S$. For higher order phase transitions, where G is smooth (but some higher order derivative is discontinuous), there is no latent heat since S is continuous. However, the specific heat $C \sim \frac{\mathrm{d}S}{\mathrm{d}T}$ diverges at the critical point.

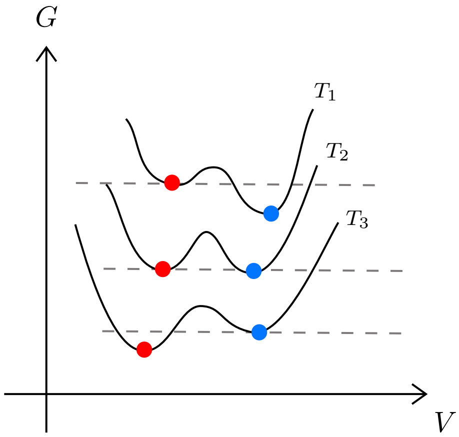


Figure 5: G vs. V plots for different temperatures.

From the previous analysis, we saw that in some interval, we had three values of volume for each pressure (out of which one was unphysical). A physical scenario to imagine is when suppose we have water above its boiling point (so it is in vapour phase). The Gibbs potential has its minima at two volumes as shown in Fig.5 for curve at T_1. The system takes the minima which is lower (the right one).

As we decrease the temperature, at some temperature, both the minima become equal, as for T_2. This is the *critical temperature.* As temperature is reduced further, the other minima becomes the more stable one. The system was in the right minima which was till now the *stable* one. It still remains in the right one, however, it now becomes *metastable.* In a very small timescale, due to some fluctuations, a *nucleus* of condensed liquid (the stable state) forms which grows rapidly and the entire system undergoes the transition.

3.2. Grand canonical basis of phase coexistence

Having seen the physical picture, we will discuss how the liquid-gas phase transition is brought about by considering the partition function that we had found out earlier from Eq(1). However, we used the uniform density approximation for this which is not entirely correct at phase coexistence since the gas (low density) and liquid phases (high density) have weirdly different densities (however, each phase can be assumed to have a uniform density separately).

This problem can be circumvented by considering the *grand canonical ensemble* where we fix the chemical potential μ and the number of particles (and in turn, the density) is appropriately adjusted for different phases. The grand partition function Ξ is defined as,

$$\Xi(\mu, T, V) = \sum_{N=0}^{\infty} e^{\beta\mu N} \mathcal{Z}(N, T, V) \tag{8}$$

where $\mathcal{Z}$ is the canonical partition function. We can appropriately modify the canonical partition function to write is as an expontential:

$$\begin{aligned}
e^{\beta\mu N + \ln \mathcal{Z}} &= \exp\left[\beta\mu N\right] \cdot \exp\left[\frac{\beta a N^2}{V}\right] \cdot \exp\left[N \ln\left(V - \frac{N\omega}{2}\right)\right] \cdot \exp\left[-\ln N! - N \ln \lambda_{\mathrm{T}}^3\right] \\
&= \exp\left[\beta\mu N\right] \cdot \exp\left[\frac{\beta a N^2}{V}\right] \cdot \exp\left[N \ln N + N \ln\left(\frac{V}{N} - \frac{\omega}{2}\right)\right] \cdot \exp\left[-\ln N! - N \ln \lambda_{\mathrm{T}}^3\right] \\
&= \exp\left[\beta\mu N + \frac{\beta a N^2}{V} + N \ln N + N \ln\left(\frac{V}{N} - \frac{\omega}{2}\right) - \ln N! - N \ln \lambda_{\mathrm{T}}^3\right] \\
&= \exp\left[\beta\mu N + \frac{\beta a N^2}{V} + \cancel{N \ln N} + N \ln\left(\frac{V}{N} - \frac{\omega}{2}\right) - \cancel{N \ln N} + N - N \ln \lambda_{\mathrm{T}}^3\right] \\
&= \exp\left[\frac{\beta a N^2}{V} + N \ln\left(\frac{V}{N} - \frac{\omega}{2}\right) + N\Delta\right]
\end{aligned}$$

where we have used Sterling's approximation, $\ln N! \approx N\ln N - N$ and $\Delta := (1+\beta\mu - \ln\lambda_{\rm T}^3)$. We can now use the saddle-point approximation to evaluate the sum, which gives only a significate contribution for some $N = N^*$.

$$\Xi \approx \exp\left[\frac{\beta a N^{*2}}{V} + N^*\ln\left(\frac{V}{N^*} - \frac{\omega}{2}\right) + N^*\Delta\right] \tag{9}$$

From appendix A, the grand potential is,

$$-PV = \Phi = -k_{\rm B}T\ln\Xi \implies \beta P = \frac{\ln\Xi}{V} \approx \left[\frac{\beta a N^{*2}}{V^2} + \frac{N^*}{V}\ln\left(\frac{V}{N^*} - \frac{\omega}{2}\right) + \frac{N^*}{V}\Delta\right]$$

We define $n^* = \frac{N^*}{V}$, where n^* is the solution of the equation $\frac{\partial\Psi}{\partial n} = 0$ where $\Psi(n)$ is defined as,

$$\Psi(n) = \beta a n^2 + n\ln\left(\frac{1}{n} - \frac{\omega}{2}\right) + n\Delta$$

Carrying out the differentiation $\frac{\partial\Psi}{\partial n}$, we obtain

$$2\beta a n + \ln\left(\frac{1}{n} - \frac{\omega}{2}\right) + \frac{n}{\frac{1}{n} - \frac{\omega}{2}} \times \left(-\frac{1}{n^2}\right) + \Delta = 0 \implies \boxed{\Delta = \frac{1}{1 - \frac{\omega n}{2}} - 2\beta a n - \ln\left(\frac{1}{n} - \frac{\omega}{2}\right)} \tag{10}$$

Substituting this value of Δ in $\Psi(n)$ we get,

$$\Psi(n^*) = \beta a N^{*2} + \cancel{n^*\ln\left(\frac{1}{N^*} - \frac{\omega}{2}\right)} + \frac{n^*}{1 - \frac{n^*\omega}{2}} - 2\beta a n^{*2} - \cancel{n^*\ln\left(\frac{1}{n^*} - \frac{\omega}{2}\right)} = \frac{1}{\frac{1}{n^*} - \frac{\omega}{2}} - \beta a N^{*2}$$

Comparing this with Van der Waal's equation, this gives us $P = P_{\rm can.}$ where $P_{\rm can.}$ is the canonical ensemble pressure. This tells us that the grand canonical and the canonical pressure are identical at particular value of the density n^*. Here we have assumed only a unique n^* which maximises the sum. However, the sum in Eq (9) can be peaked at multiple different values of densities and the appropriate uniform density is chosen for that value which maximises $\psi(n)$.

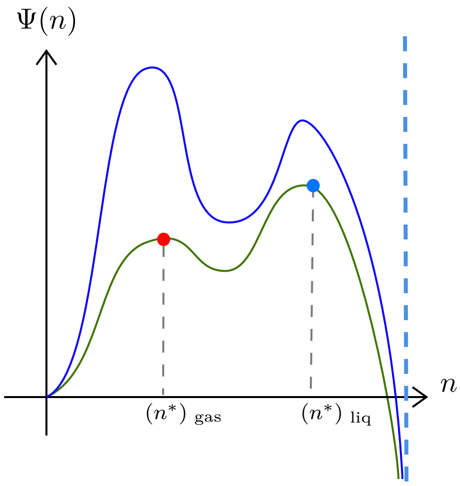


Figure 6: Plot of $\Psi(n)$ vs. n at different temperature. Green plot is for $T < T_c$ and blue is for $T > Tc$. The lower density corresponds to gas and higher density correpsonds to liquid. At $n = \frac{2}{\omega}$, $\Psi(n) \to -\infty$ which has been shown by the blue vertical dashed line.

In the above figure, the blue curve denotes $T > T_c$ and the green curve denotes $T < T_c$. For both $T < T_c$ and $T > T_c$, we have two peaks in the $\Psi(n)$ vs. n plot, however, only the right peak is chosen for $T < T_c$ (corresponding to liquid density) and the left peak is chosen for $T > T_c$ (corresponding to gas density). Considering contribution from both peaks,

$$\ln\Xi \approx \ln\left(e^{V\Psi(n^*_{\rm liq})} + e^{V\Psi(n^*_{\rm gas})}\right) = V\Psi(n^*_{\rm liq}) + \ln\left(1 + e^{V(\Psi(n^*_{\rm gas}) - \Psi(n^*_{\rm liq}))}\right)$$

In the thermodynamic limit, if $T > T_c$, $\Psi(n^*_{\rm gas}) > \Psi(n^*_{\rm liq})$ then $\ln\Xi \approx V\Psi(n^*_{\rm gas})$ and hence the gas phase dominates. For $T < T_c$, $\Psi(n^*_{\rm gas}) < \Psi(n^*_{\rm liq})$ and hence the liquid phase dominates. From this, we can say that only in the thermodynamic limit, there is a *singularity* in the density at $T = T_c$ (there are no phase transitions in finite systems)!

Lecture 04: Yang-Lee Theory for Phase Transition

We will now discuss about a mathematical basis of condensation based on Yang and Lee's theory of phase transition [1] which is based on the analysis of the zeroes of the grand partition function. The partition function seems to be an analytic function since it essentially consists of sums of well-behaved exponentials.

However, phase transitions are characterised by non-analyticity which leads us to think about how the analytic partition function will capture the essence of the non-analytic phase transition. As discussed before, *true* phase transitions occur only in the thermodynamic limit and in the thermodynamic limit, partition function can indeed become non-analytic (since the limit functions of a sequence of analytic function need not be analytic).

4.1. Singularities and Phase Transition

We will write Eq (8) in terms of the *fugacity* $z = e^{\beta\mu}$ and also consider a finite system with volume V (and then take ultimately take $V \to \infty$). Only a finite number of particles, say M, can be crammed into this finite volume, hence

$$\Xi(z, V) = \sum_{N=0}^{M} z^N \mathcal{Z}(N, V, T) = 1 + z\mathcal{Z}_1 + z^2\mathcal{Z}_2 + \ldots + z^M \mathcal{Z}_M \tag{11}$$

Note that the above equation is a well-behaved polynomial of degree M in z and its coefficients $\mathcal{Z}_N$ are all positive which implies that $\Xi(z, V)$ has no real positive root. From the grand potential $\Phi = -PV = -k_{\mathrm{B}}T \ln \Xi$, we can write the pressure P and density ρ in the thermodynamic limit,

$$\begin{aligned} P &= \lim_{V\to\infty} \frac{k_{\mathrm{B}}T}{V} \ln \Xi(z, V) \\ \langle N \rangle &= z\frac{\partial \ln \Xi}{\partial z} \implies \rho = \lim_{V\to\infty} \frac{\langle N \rangle}{V} = \lim_{V\to\infty} \frac{z}{V} \frac{\partial \ln \Xi(z, V)}{\partial z} \end{aligned} \tag{12}$$

These quantities will not have any singularity unless $\Xi(z, V) = 0$ which cannot hold true, unless we cosider a *complex* fugacity. Then for finite V, the algebraic equation $\Xi = 0$ has M roots appearing in complex-conjugate pairs. As V increases, the number of roots also increases (linearly with V) and moves around in the complex plane, excluding the real axis. In the limit $V \to \infty$, some of these roots touch the real axis and this describes a phase transition. With regard to this, we discuss two theorems as stated in the original paper.

Theorem 1:
For all positive values of z, the limit

$$L = \lim_{V\to\infty} \frac{1}{V} \ln \Xi(z, V)$$

exists and is independent of the shape of the volume holding the particles, with the assumption that the surface area does not scale faster than $V^{2/3}$.

Moreover, $L(z)$ is a continuous, monotonically increasing function of z.

Theorem 2:
Suppose in the complex z plane, there exists a region $\mathcal{R}$ containing the positive real axis, which has no roots of Ξ, then $\lim_{V\to\infty} \frac{1}{V} \frac{\partial^k \ln \Xi}{\partial z^k}$ is uniformly convergent and the limit function is analytic for all $z \in \mathcal{R}$.

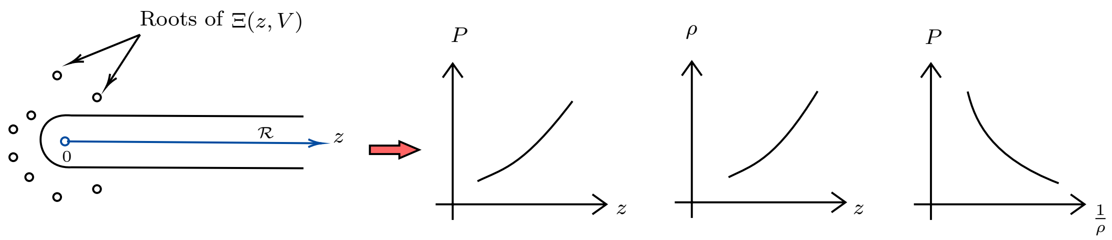
Moreover, the operations $\lim_{V\to\infty}$ and $z\frac{\partial}{\partial z}$ can be interchanged.

The above theorem implies that,

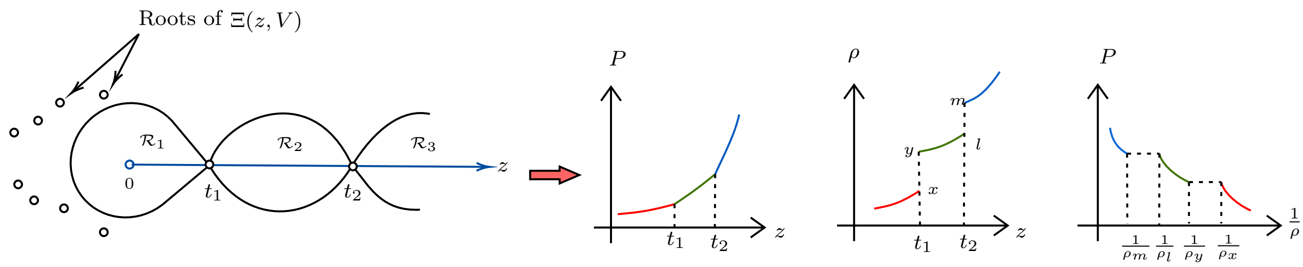
$$\rho = \lim_{V\to\infty} \frac{z}{V}\frac{\partial \ln \Xi}{\partial z} \iff z\frac{\partial}{\partial z}\left(\lim_{V\to\infty}\frac{1}{V}\ln \Xi\right) = z\frac{\partial(\beta P)}{\partial z} \tag{13}$$

If such a region $\mathcal{R}$ does not exist, then the limit does not exist which makes sense physically, since the density ρ as defined in Eq (12), does not assume a single value at the condensation point. Hence, everything depends on the existence of $\mathcal{R}$

- As $V \to \infty$, if a region $\mathcal{R}$ exists which fully encloses the positive real axis, then P exists from Thm (1) and is an analytic, increasing function of z. From Thm (2), ρ also exists and is an analytic function of z. It can be shown that ρ is also increasing with z. P and ρ are related by Eq (13). The system under consideration is in a *single phase*.



- On the other hand, if the roots of $\Xi(z,V)$ approach some specific values $z = t_1, t_2 \ldots$ on the real axis as $V \to \infty$, from the figure below, there will exist regions $\mathcal{R}_1, \mathcal{R}_2 \ldots$ within which Thm (2) will be satsified piecewise and hence, within these regions, the system exists in a single phase as argued before. At the points $t_1, t_2 \ldots$, the pressure P will be continuous from Thm (1), but ρ might have some non-analyticity since this does not satisfy the condition for Thm (2). The system now possess *multiple phases*.



It is to be noted that the discontinuity in the density is only possible in the thermodynamic limit. For finite system, we will get a *continuous crossover* due to *finite-size effects*.

If $\frac{\partial P}{\partial z}$ is discontinuous, then we have a *first-order phase transition* as shown in the figure above. If $\frac{\partial P}{\partial z}$ is continuous but $\frac{\partial^2 P}{\partial z^2}$ is discontinuous, then we obtain a *second-order phase transition*.

From this discussion, we observe that the study of phase transitions can be reduced to just studying the distribution of the zeroes of the grand partition function.

Lecture 05: More on Van der Waal's Equation

In the previous section, we saw a mathematical theory of condensation based on the zeroes of the grand partition function. Let us now move back to the Van der Waal's equation and try to analyse the critical point.

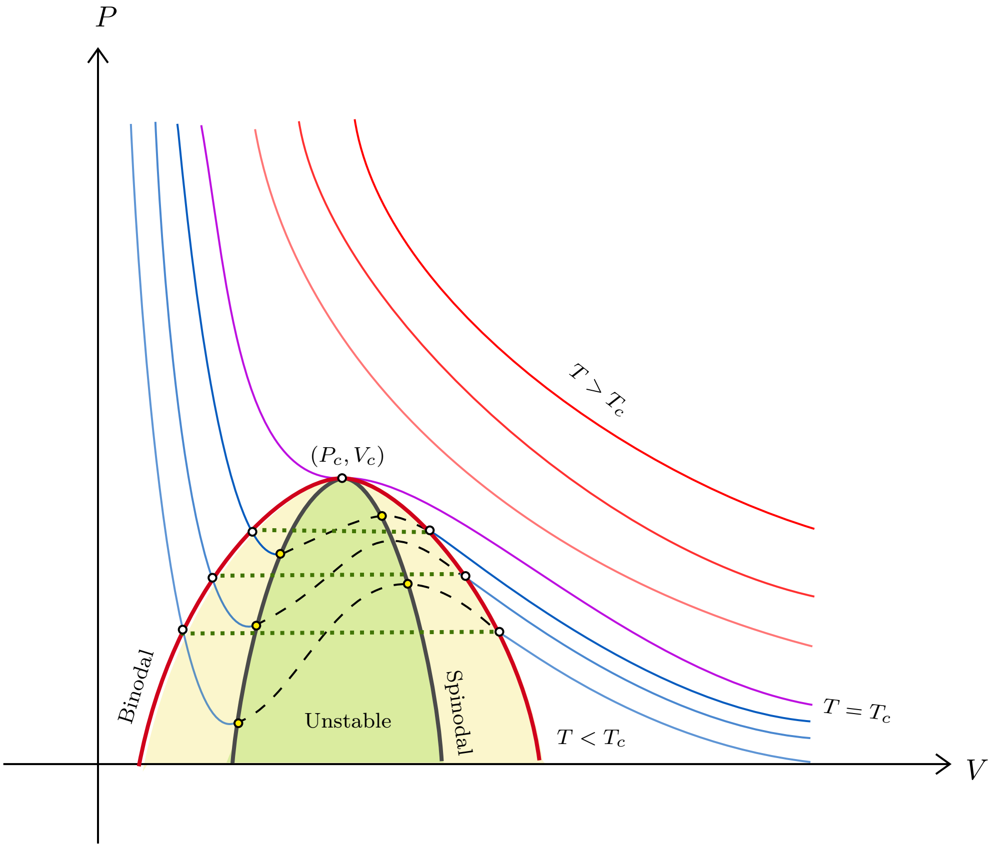


Figure 7: Van der Waal's isotherm. The figure shows Van der Waal's isotherms for $T < T_c$ ($T > T_c$) with blue (red) lines. The curve for $T = T_c$ is shown with purple line. Maxwell's construction is shown with green dotted line. The red line denotes the *binodal line* and the gray line denotes the *spinodal* line. The yellow region denote the metastable state while the green region denote the unstable states.

In Fig.??, we see the Van der Waal's isotherms at different temperature regime. There exists a temperature T_c such that at a particular point (V_c, P_c), the isotherm has zero slope and zero curvature. Thus the critical point is characterised by

$$\frac{\partial P}{\partial V} = 0 \qquad \frac{\partial^2 P}{\partial V^2} = 0$$

Defining $v = \frac{V}{N}$, from Van der Waal's equation we can write,

$$P = \frac{k_B T}{v-b} - \frac{a}{v^2} \implies \frac{\partial P}{\partial V} = -\frac{k_B T}{(v-b)^2} + \frac{2a}{v^3} \implies \frac{\partial^2 P}{\partial V^2} = \frac{2k_B T}{(v-b)^3} - \frac{6a}{v^4} \tag{14}$$

Equating the derivative to zero, we get $2a(v_c - b)^2 = k_B T v_c^3$ and equating the second derivative to zero, we get $3a(v_c - b)^3 = k_B T v_c^4$. Dividing both these equations, we get

$$\frac{3}{2}(v_c - b) = v_c \implies 3(v_c - b) = 2v_c \implies v_c = 3b$$

Substituting this in any one of the above equation we get $k_B T_c = \frac{8a}{27b}$ and substituting v_c and T_c in Van der Waal's equation, we get $P_c = \frac{a}{27b^2}$. Thus our critical parameters are,

$$\boxed{P_c = \frac{a}{27b^2} \qquad v_c = 3b \qquad k_B T_c = \frac{8a}{27b}}$$

5.1. Approaching Universality

The Van der Waal's equation depends on the microscopic details of the system, a and b. These vary for different gases and hence, the equation cannot describe universal properties. However, let us define the *reduced parameters*,

$$\Pi = \frac{P}{P_c} \qquad \nu = \frac{v}{v_c} \qquad \tau = \frac{T}{T_c}$$

We now put these in the equation of state, which gives us

$$\begin{aligned}\left(\Pi P_c + \frac{a}{\nu^2 v_c^2}\right)(\nu v_c - b) = k_B \tau T_c \implies & \left(\frac{\Pi a}{27b^2} + \frac{a}{9\nu^2 b^2}\right)(3b\nu - b) = \frac{8a\tau}{27b} \\ \implies & \frac{3ab}{27b^2}\left(\Pi + \frac{3}{\nu^2}\right)\left(\nu - \frac{1}{3}\right) = \frac{8a\tau}{27b} \\ \implies & \boxed{\left(\Pi + \frac{3}{\nu^2}\right)\left(\nu - \frac{1}{3}\right) = \frac{8\tau}{3}}\end{aligned} \tag{15}$$

The above dimensionless Van der Waal's equation in terms of the reduced parameters, is independent of a and b and is thus *universal*, in the sense that, different gases are expected to follow the same equation. The generalisation is known as the *principle of corresponding states*, which says that all fluids when compared with respect to reduced parameters, have the same *compressibility factor*. The compressibility factor is defined as $Z = \frac{Pv}{k_B T}$, with the critical value predicted by Van der Waal to be,

$$Z_c = \frac{P_c v_c}{k_B T_c} = \frac{3ab}{27b^2 \times \frac{8a}{27b}} = \frac{3}{8} = 0.375$$

Experimentally, the critical value of the compressibility factor is found to be in the range $0.28 - 0.33$, thus, Van der Waal's equation does not provide an accurate quantitative estimate.

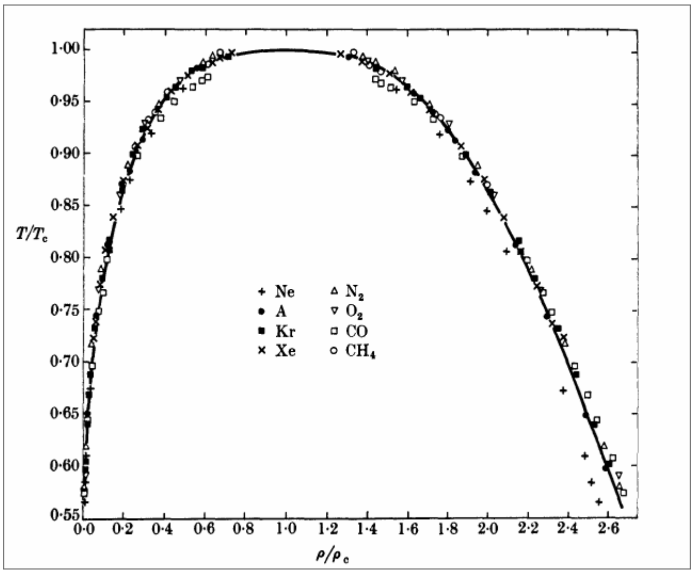


Figure 8: The Guggenheim plot for different gases with reduced parameter, showing that the trend follows a universal function (Source: Guggenheim (1945) [2]).

5.2. Critical Opalescence

Critical opalescence refers to an enhanced scattering of light by a substance in this critical region. Such enhanced scattering renders the substance milky white in reflected light.

As we know, at the critical point, $\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T \to \infty$ since the slope of the isotherm is zero at the critical point. As $\kappa_T \propto \sigma_N$ where σ_N is the number fluctuations in the system, there is a huge number fluctuation, due to which the scattering of light is intensified.

5.3. Critical Point Behaviour

We will now discuss the behaviour of gases in the vicinity of the critical point to analyse universality. For $T > T_c$, there is always a unique volume corresponding to each pressure and hence, we can apply

Taylor expansion around v_c to write the pressure as,

$$P(T,v) = P(T,v_c) + \left.\frac{\partial P}{\partial v}\right|_T (v-v_c) + \frac{1}{2}\left.\frac{\partial^2 P}{\partial v^2}\right|_T (v-v_c)^2 + \frac{1}{6}\left.\frac{\partial^3 P}{\partial v^3}\right|_T (v-v_c)^3 + \mathcal{O}\left[\left(\frac{v-v_c}{v_c}\right)^4\right]$$

Each of the coefficients are functions of temperature and hence, can be Taylor expanded about T_c.

$$\begin{aligned}
P(T,v_c) &= \underbrace{P(T_c,v_c)}_{P_c} + \alpha(T-T_c) + \mathcal{O}\left[\left(\frac{T-T_c}{T_c}\right)^2\right] \\
\left.\frac{\partial P}{\partial v}\right|_{T,v_c} &= \left.\cancelto{0}{\frac{\partial P}{\partial v}}\right|_{T_c,v_c} - a(T-T_c) + \mathcal{O}\left[\left(\frac{T-T_c}{v_c}\right)^2\right] \\
\left.\frac{\partial^2 P}{\partial v^2}\right|_{T,v_c} &= \left.\cancelto{0}{\frac{\partial^2 P}{\partial v^2}}\right|_{T_c,v_c} + b(T-T_c) + \mathcal{O}\left[\left(\frac{T-T_c}{T_c}\right)^2\right] \\
\left.\frac{\partial^3 P}{\partial v^3}\right|_{T,v_c} &= -c + \mathcal{O}\left[\left(\frac{T-T_c}{T_c}\right)^2\right]
\end{aligned}$$

At the critical point the slope and the curvature vanishes and hence the first terms in the second and third equation is zero. Using the stability condition in Eq. (5), $a > 0$ for $T > T_c$ and $c > 0$. b can have any arbitrary sign[1]. Let us put these expression in the original Taylor expansion.

$$P(T,v) = P_c + \alpha(T-T_c) - a(T-T_c)(v-v_c) + \frac{b}{2}(T-T_c)(v-v_c)^2 - \frac{c}{6}(v-v_c)^3 + \text{higher orders}\ldots \tag{16}$$

From this, we can predict the following things about different quantities:

- Let us consider only upto linear order in $(T - T - c)$ and $(v - v_c)$, hence we neglect the fourth, fifth and higher order terms. This gives us an approximate analytic solution for the pressure,

  $$\boxed{P(T,v) = P_c + \alpha(T-T_c) - a(T-T_c)(v-v_c)}$$

  We obtain $\frac{\partial P}{\partial v} = -a(T-T_c)$ and hence the isothermal compressibility at v_c scales as,

  $$\lim_{T\to T_c^+} \kappa(T,v_c) = -\frac{1}{v_c}\frac{\partial v}{\partial P} \sim \frac{1}{T-T_c}$$

  Experimental results identify the exponent as $\gamma \approx 1.3$ where $\kappa(T,v_c) \sim (T-T_c)^{-\gamma}$.

- At $T = T_c$, the isotherm behaves as

  $$P = P_c - \frac{c}{6}(v-v_c)^3 \implies (P-P_c) \sim (v-v_c)^3$$

  Experimental results identify the exponent as $\delta \approx 5.0$ where $(P-P_c) \sim (v-v_c)^\delta$.

The above analysis cannot be applied for $T < T_c$ since then for each pressure, we have two volumes and Taylor expansion cannot be effectively applied. Let us do the analysis in an alternate way using Eq. (15). Consider the two volumes v_{liq} and v_{gas} for which the pressure is same at say some temperature T. Then, from the equation we can write,

$$\Pi = \frac{8\tau/3}{\nu_g - 1/3} - \frac{3}{\nu_g^2} = \frac{8\tau/3}{\nu_l - 1/3} - \frac{3}{\nu_l^2}$$

[1]These a, b are not the ones appearing in the Van der Waal's equation.

From this we obtain,

$$8\tau\left[\frac{1}{\nu_g-1}-\frac{1}{\nu_l-1}\right]=3\left[\frac{1}{\nu_l^2}-\frac{1}{\nu_g^2}\right]=3\left[\frac{\nu_g^2-\nu_l^2}{\nu_l^2\nu_g^2}\right]\implies\boxed{\tau=\frac{(3\nu_l-1)(3\nu_g-1)(\nu_l+\nu_g)}{8\nu_g^2\nu_l^2}}$$

Note that the equation is symmetric in ν_g and ν_l and $\nu_g=\nu_l=1$ at the critical point, which implies that we can write

$$\nu_g=1+\frac{\varepsilon}{2}\qquad\nu_l=1-\frac{\varepsilon}{2}$$

where $\varepsilon\equiv\nu_g-\nu_l$. We substitute this in the expression for τ, leading us to

$$\tau=\frac{2\big(2-\frac{3\varepsilon}{2}\big)\big(2+\frac{3\varepsilon}{2}\big)}{8\big[\big(1+\frac{\varepsilon}{2}\big)\big(1-\frac{\varepsilon}{2}\big)\big]^2}=\frac{\big(4-\frac{9\varepsilon^2}{4}\big)}{4\big(1-\frac{\varepsilon^2}{4}\big)^2}=\left(1-\frac{9\varepsilon^2}{16}\right)\left(1-\frac{\varepsilon^2}{4}\right)^{-2}\approx\left(1-\frac{9\varepsilon^2}{16}\right)\left(1+\frac{\varepsilon^2}{2}\right)=1-\frac{\varepsilon^2}{16}+\mathcal{O}(\varepsilon^4)$$

Then substituting for the real variables, we have

$$(v_{\text{gas}}-v_{\text{liq}})\sim(T_c-T)^{1/2}$$

Experimental results predict an exponent $\beta\approx0.3$ where $(v_{\text{gas}}-v_{\text{liq}})\sim(T_c-T)^{\beta}$.

Van der Waal's equation thus does not provide a perfect quantitative description of condensation, however, qualitative descriptions are more or less accurate. The exponents $\beta,\gamma,\delta\ldots$ are called *critical exponents* and many people dedicate their lives in finding the numerical value of these exponents.

Appendices

From the first law we have the energy conservation equation,

$$\mathrm{d}U=T\mathrm{d}S-P\mathrm{d}V+\mu\mathrm{d}N$$

and thus the internal energy is a function of S,V and N. We know that the internal energy is a homogenous function of degree 1, on physical grounds, since if we scale the system variables by λ, U is also appropriately scaled (extensitivity)

$$U(\lambda S,\lambda V,\lambda N)=\lambda U(S,V,N)$$

Now, from Euler's theorem of homogenous functions, we know that any homogenous function f of degree k can be written as,

$$kf(x_1,x_2\ldots x_n)=\sum_{i=1}^{n}x_i\frac{\partial f}{\partial x_i}$$

Using this theorem for U, we have

$$U=S\frac{\partial U}{\partial S}+V\frac{\partial U}{\partial V}+N\frac{\partial U}{\partial N}$$

Substituting the values from the first law, we obtain $U=TS-PV+\mu N$

A. Legendre Transformation

From classical mechanics, we know that the Legendre transformation is used to convert Lagrangian $L(x,\dot{x})$ to the Hamiltonian $H(x,p)$ and vice versa. Note how we switch $\dot{x}$ in L for p in H. We will call the variable that we want to switch to as the 'conjugate' of the variable we are switching. The Legendre transformation of the Lagrangian gives us the Hamiltonian,

$$H(x,p)=p\dot{x}-L$$

A function $f(x)$ with the Legendre transform $f^*(p)$ satsifies $f(x) + f^*(p) = xp$. We will use this for thermodynamic quantities also, however we will use an alternate definition using minus sign, $f(x) - f^*(p) = xp$. From the first law, we know that,

$$U = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N$$

As clearly seen, the conjugate variable pairs are (S, T), $(V, -P)$, (N, μ). We will consider the starting point as the internal energy $U(S, V, N)$ which is a function of S, V, N from the statement of the first law. However, in any experiment, keeping these as control parameters are way too hard. Think about this, controlling entropy of a system is an arduous task but controlling its conjugate variable, T, is easy-peasy.

We get four different 'potentials' by just Legendre transformation (which is basically multiplying the conjugate variables together and subtracting from the function) of the internal energy, switching between the different conjugate variables.

- **Helmholtz Free Energy**: switching $S \leftrightarrow T$

$$F(T, V, N) = U - TS \implies \mathrm{d}F = -S\mathrm{d}T - P\mathrm{d}V + \mu\mathrm{d}N$$

- **Enthalpy**: switching $V \leftrightarrow -P$

$$H(S, P, N) = U + PV \implies \mathrm{d}H = T\mathrm{d}S + V\mathrm{d}P + \mu\mathrm{d}N$$

- **Gibb's Free Energy**: switching both $S \leftrightarrow T$ and $V \leftrightarrow -P$

$$G(T, P, N) = U - TS + PV \implies \mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}P + \mu\mathrm{d}N$$

- **Grand Potential**: switching both $S \leftrightarrow T$ and $N \leftrightarrow \mu$

$$\Phi(T, V, \mu) = U - TS - \mu N = -PV \implies \mathrm{d}\Phi = -S\mathrm{d}T - P\mathrm{d}V - N\mathrm{d}\mu$$

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